A

Project Phase-II Report on

"GRADX.AI
AI-Powered Semantic Answer Evaluation System with Advanced Anti-Cheating Capabilities"

Submitted To

Shivaji University, Kolhapur In Fulfilment of

the Requirement

For the Degree Of

BACHELOR OF TECHNOLOGY IN

(COMPUTER SCIENCE AND ENGINEERING)

Submitted by

PRN	Name of Student
2223000839	Mr. Kunal Shinde
2223000529	Mr. Tushar Sutar
2223000279	Mr. Vinayak Jadhav
2223000649	Mr. Vaibhav Sutar
2223000861	Mr. Pranav Telvekar

Under the guidance of **Prof. Ajay Kapse**

Submitted at

Kolhapur Institute of Technology's College of Engineering Kolhapur (Empowered Autonomous)

Year 2025-2026

CERTIFICATE

This is to certify that **Mr. Kunal Shinde (PRN: 2223000839)**, **Mr. Tushar Sutar (PRN: 2223000529)**, **Mr. Vinayak Jadhav (PRN: 2223000279)**, **Mr. Vaibhav Sutar (PRN: 2223000649)**, and **Mr. Pranav Telvekar (PRN: 2223000861)** have successfully completed the project entitled **"GRADX: AI-Powered Semantic Answer Evaluation System with Advanced Anti-Cheating Capabilities"** in partial fulfilment of the requirements for the award of the degree of **Bachelor of Technology in Computer Science and Engineering** at **Kolhapur Institute of Technology's College of Engineering, Kolhapur (Empowered Autonomous)** during the academic year **2025-26**.

Date: February 14, 2026

Place: **Kolhapur Institute of Technology's College of Engineering Kolhapur**

(Empowered Autonomous)

Prof. Ajay Kapse		**Dr. L.A. Hadimani**
Project Guide	**External Examiner**	**Head CSE**

Dr. M. B. Vanarotti
Director

DECLARATION

We hereby declare that the project entitled **"GRADX: AI-Powered Semantic Answer Evaluation System with Advanced Anti-Cheating Capabilities"** submitted to **Kolhapur Institute of Technology's College of Engineering, Kolhapur (Empowered Autonomous), Maharashtra, INDIA**, in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in Computer Science and Engineering**, is our own bonafide work.
We further declare that the work presented in this project has not been submitted to any other university or institution for the award of any degree or diploma.

Date: February 14, 2026

Place: **Kolhapur Institute of Technology's College of Engineering Kolhapur (Empowered Autonomous)**

Mr. Kunal Shinde

Mr. Tushar Sutar

Mr. Vinayak Jadhav

Mr. Vaibhav Sutar

Mr. Pranav Telvekar

ACKNOWLEDGEMENT

We would like to express our deep gratitude towards **Ajay Kapse Sir** and **Dr. L.A. Hadimani, Head of Department**, for their constant encouragement and guidance. Their support and cooperation throughout the project work has been of immense help to us.We express our sincere thanks to all the teaching and non-teaching staff and all those who have directly or indirectly helped in making the project a success.

Yours Sincerely,

Mr. Kunal Shinde	2223000839
Mr. Tushar Sutar	2223000529
Mr. Vinayak Jadhav	2223000279
Mr. Vaibhav Sutar	2223000649
Mr. Pranav Telvekar	2223000861

TABLE OF CONTENT

Title	Page no.
Certificate	
Declaration	
Acknowledgement	
Abstraction	**6**
Introduction	**7**
Problem Statement	**8**
Objectives	**9**
Scope of the Project	**10**
Literature Survey	**11**
Existing System	**12**
Proposed System	**13**
Feasibility Study	**14**
System Architecture / Block Diagram	**15**
Methodology	**16**
Work Plan / Activity Schedule	**17**
Progress Work Completed (Phase-II)	**18**
Expected Outcomes	**19**
Future Enhancements	**20**
Conclusion & References	**21**

ABSTRACT

GradX is an advanced AI-powered answer evaluation system designed to revolutionize academic assessment through semantic analysis, contextual understanding, and intelligent feedback generation. The system integrates state-of-the-art natural language processing, machine learning models, and anti-cheating mechanisms to create a comprehensive evaluation ecosystem. By automating answer assessment while maintaining semantic accuracy, GradX significantly reduces evaluation time, eliminates human bias, and provides detailed constructive feedback. The project aims to deliver a scalable, intelligent, and reliable solution capable of transforming educational assessment through AI-driven insights.

1. INTRODUCTION

Traditional answer evaluation systems in education often struggle with subjectivity, inconsistency, and time-consuming manual processes. As academic institutions face increasing student volumes and the need for standardized assessment, the limitations of conventional evaluation methods become more apparent. The rise of online education and digital submissions further exacerbates these challenges, creating an urgent need for automated, yet intelligent evaluation systems.

GradX bridges these gaps by seamlessly integrating semantic similarity analysis, concept mapping, terminology validation, and advanced cheating detection mechanisms. Through sophisticated AI models, the system enables automatic evaluation of conceptual understanding, contextual application of knowledge, and detection of superficial learning patterns.

By ensuring consistent scoring, detailed feedback generation, and prevention of assessment manipulation, GradX significantly enhances evaluation quality while reducing faculty workload. The system creates a more efficient, transparent, and technology-empowered educational assessment environment.

2. PROBLEM STATEMENT

Existing evaluation systems, whether manual or automated, face significant challenges in assessing conceptual understanding. Manual evaluation suffers from inherent subjectivity, inconsistency across different evaluators, and is a highly time-consuming process, especially for large student populations.

Basic automated systems that rely on simple keyword matching are insufficient. They fail to capture semantic meaning and contextual understanding, making them vulnerable to gaming through "keyword stuffing," where students use relevant terms without demonstrating genuine comprehension. The absence of sophisticated cheating detection, lack of detailed feedback mechanisms, and an inability to evaluate the logical flow of an answer further limit current solutions.

There is a critical need for an intelligent evaluation system that can:

- Understand semantic meaning beyond keyword matching.
- Detect and penalize superficial learning attempts like keyword stuffing.
- Provide constructive, detailed feedback for student improvement.
- Maintain consistency across multiple evaluations and sessions.
- Scale efficiently for large student populations.

3. OBJECTIVES

Primary Objectives:

1. To develop an AI-powered system capable of semantic understanding and contextual evaluation of answers.
2. To implement advanced mechanisms for detecting keyword stuffing and evaluating the quality of explanations.
3. To build a comprehensive feedback generation system providing actionable insights for student improvement.

Secondary Objectives:

- Reduce evaluation time by 70–80% through automation.
- Achieve 98% evaluation consistency across different sessions to eliminate bias.
- Validate the quality and proper usage of technical terminology.
- Assess the logical flow and coherence of answers.
- Ensure scalability for institutional deployment while maintaining data privacy and security.

4. SCOPE OF THE PROJECT

The GradX system is applicable to:

- University and college examinations
- Online course assessments
- Competitive exam evaluations
- Corporate training assessments
- Research paper evaluation assistance

Limitations:

- Requires well-defined model answers for optimal performance
- Performance depends on question complexity and domain specificity
- May require domain-specific fine-tuning for specialized subjects
- Initial setup requires faculty configuration for each evaluation context

5. LITERATURE REVIEW

Research indicates that AI-powered evaluation systems significantly outperform traditional methods in consistency and efficiency. Semantic similarity models using transformer architectures have shown remarkable accuracy in understanding contextual meaning. Studies confirm that multi-dimensional evaluation approaches combining keyword analysis, semantic understanding, and contextual validation provide the most reliable assessment outcomes.

Key Research Areas:

- **Transformer-based Semantic Analysis**: BERT and Sentence-BERT models for understanding contextual meaning
- **Concept Mapping**: Automated identification and evaluation of key concepts in answers
- **Cheating Detection**: Pattern recognition for identifying keyword stuffing and superficial responses
- **Feedback Generation**: AI-driven constructive feedback generation for learning improvement
- **Multi-modal Evaluation**: Combining multiple evaluation dimensions for comprehensive assessment

6. EXISTING SYSTEM

Traditional Evaluation Systems:
Current evaluation methodologies, whether manual or basic automated systems, face several critical limitations:
Manual Evaluation Challenges:

- Subjectivity and inconsistency across different evaluators
- Time-consuming process, especially for large classes
- Fatigue-induced errors in extended evaluation sessions
- Lack of detailed, standardized feedback mechanisms
- Inability to maintain consistent evaluation standards over time

Basic Automated Systems Limitations:

- Reliance on simple keyword matching without semantic understanding
- Inability to detect contextual application of concepts
- Vulnerability to gaming through keyword stuffing
- Lack of sophisticated cheating detection mechanisms
- Limited feedback generation capabilities
- Poor handling of paraphrased but correct answers

These limitations result in inaccurate assessments, frustrated students, and increased faculty workload, ultimately compromising the educational assessment quality.

7. PROPOSED SYSTEM

System Architecture Components:

1. Semantic Understanding Engine
- **Sentence-BERT Integration**: Advanced transformer models for semantic similarity analysis
- **Contextual Analysis**: Evaluation of answer coherence and logical flow
- **Concept Relationship Mapping**: Understanding how concepts interrelate in answers

2. Advanced Cheating Detection Module
- **Keyword Density Analysis**: Detection of excessive keyword usage
- **Sentence Structure Evaluation**: Identification of unnatural answer patterns
- **Context Validation**: Ensuring keywords are used in proper context
- **Pattern Recognition**: Identifying memorization without understanding

3. Multi-dimensional Evaluation Framework
- **Concept Coverage Analysis**: Percentage-based concept understanding measurement
- **Terminology Usage Quality**: Assessment of proper technical term usage
- **Semantic Depth Scoring**: Evaluation of answer comprehensiveness
- **Contextual Application**: Measurement of practical concept application

4. Intelligent Feedback Generation
- **Strength Identification**: Recognition of well-explained concepts
- **Improvement Areas**: Specific guidance on conceptual gaps
- **Warning System**: Alerts for potential cheating attempts
- **Personalized Suggestions**: Tailored recommendations for improvement

8. FEASIBILITY STUDY

Technical Feasibility:

- All required technologies (Sentence-BERT, spaCy, scikit-learn) are mature and well- documented
- Cloud infrastructure supports scalable deployment
- API-based architecture allows easy integration with existing systems
- Open-source components reduce licensing costs

Operational Feasibility:

- Intuitive faculty configuration interface
- Minimal training requirements for evaluators
- Seamless integration with existing educational workflows
- Comprehensive documentation and support systems

Economic Feasibility:

- Significant reduction in faculty evaluation time (70-80% time savings)
- Elimination of evaluation inconsistencies reducing re-evaluation costs
- Scalable cloud deployment reducing infrastructure costs
- Open-source technology stack minimizing licensing expenses

Legal Feasibility:

- Compliance with educational data privacy regulations
- Transparent evaluation methodology
- Configurable evaluation criteria meeting institutional standards

9. SYSTEM ARCHITECTURE / BLOCK DIAGRAM

Data Flow Explanation:

1. **Faculty Configuration**: Defines evaluation criteria, concepts, and required terminology
2. **Answer Processing**: Student answers are processed through multiple AI models
3. **Multi-dimensional Analysis**: Simultaneous evaluation of semantic understanding, concept coverage, and terminology usage
4. **Cheating Detection**: Advanced algorithms identify suspicious answer patterns
5. **Score Calculation**: Weighted scoring based on question type and evaluation criteria
6. **Feedback Generation**: Constructive feedback highlighting strengths and improvement areas
7. **Results Compilation**: Comprehensive evaluation report with detailed analytics

10. METHODOLOGY

Development Approach: Agile Methodology Phase

1: Requirements Analysis

- Conduct comprehensive stakeholder analysis
- Define functional and non-functional requirements
- Analyze existing evaluation system limitations
- Establish success metrics and evaluation criteria

Phase 2: System Design

- Architecture design and technology selection
- Database schema design for concept storage
- API design for system integration
- User interface design for faculty configuration

Phase 3: Core Engine Development

- Implement semantic similarity algorithms
- Develop concept mapping and evaluation modules
- Build cheating detection mechanisms
- Create feedback generation systems

Phase 4: Integration & Testing

- Integrate all system components
- Conduct unit testing and integration testing
- Perform performance and scalability testing
- Validate accuracy with sample evaluations

Phase 5: Deployment & Validation

- Pilot deployment in controlled environment
- Collect feedback from faculty and students
- Fine-tune algorithms based on real-world usage
- Prepare documentation and training materials

11. WORK PLAN / ACTIVITY SCHEDULE

Phase 1: Foundation (Weeks 1-4)

- Literature survey and technology research
- Requirements gathering and analysis
- System architecture design
- Development environment setup

Phase 2: Core Development (Weeks 5-12)

- Semantic analysis engine development
- Cheating detection module implementation
- Evaluation algorithm development
- Basic user interface creation

Phase 3: Advanced Features (Weeks 13-18)

- Feedback generation system
- Advanced analytics and reporting
- System integration and testing
- Performance optimization

Phase 4: Deployment (Weeks 19-24)

- Pilot testing and validation
- Bug fixes and improvements
- Documentation preparation
- Final deployment preparation

12. PROGRESS WORK COMPLETED (PHASE–2)

Building upon the foundational work of Phase-I, significant progress has been made in Phase-II to create a functional evaluation pipeline.

Completed Milestones:

- **Handwritten Text Extraction Pipeline:** Successfully implemented and tested a pipeline using **TrOCR** for extracting text from scanned answer sheets.
- **Image Preprocessing:** Utilized **OpenCV** for denoising, grayscale conversion, and rotation correction, significantly improving OCR accuracy.
- **Layout Parsing:** Applied **LayoutParser** and **DocTR** to accurately identify and segment question and answer regions from complex scan layouts.
- **Structured Output:** The pipeline now reliably converts raw scans into organized JSON Q&A mappings.
- **Semantic Analysis Integration:** Successfully integrated **Sentence-BERT** and **BERTScore** to perform deep semantic comparison between student and model answers, marking the shift from "reading text" to "understanding meaning."
- **Advanced Evaluation Framework:** Developed methods for contextual analysis (evaluating logical flow) and terminology validation.

Current System Capabilities:

- End-to-end pipeline from scanned image to structured text.
- Semantic similarity analysis with a target accuracy of >90%.
- Framework for evaluating conceptual depth and coherence.

13. EXPECTED OUTCOMES

Technical Outcomes:

- Semantic understanding accuracy: >90%
- Cheating detection accuracy: >95%
- Evaluation consistency: 98% across multiple sessions
- Processing speed: <5 seconds per answer evaluation
- System availability: 99.5% uptime

Educational Outcomes:

- Reduction in evaluation time: 70-80%
- Elimination of evaluation bias and inconsistency
- Improved student learning through detailed feedback
- Enhanced faculty productivity and satisfaction
- Standardized assessment across institutions

Operational Outcomes:

- Real-time analytics and performance tracking
- Comprehensive reporting and assessment insights
- Easy integration with existing educational platforms
- Minimal maintenance and operational overhead

14. FUTURE ENHANCEMENTS

Short-term Enhancements:

- Multi-language support for evaluation.
- Integration with popular Learning Management Systems (LMS) like Moodle.
- Advanced analytics dashboard for faculty.

Medium-term Enhancements:

- Plagiarism detection integration.
- Support for evaluating diagrams and flowcharts.
- Predictive analytics to identify at-risk students.

Long-term Vision:

- AI-powered question generation based on learning outcomes.
- Cross-institutional benchmarking and standardization.
- Global educational assessment standards contribution.

15. CONCLUSION

GradX represents a significant advancement in educational assessment technology, combining sophisticated AI capabilities with practical educational needs. By addressing the fundamental limitations of traditional evaluation systems while incorporating advanced cheating detection and detailed feedback generation, GradX sets a new standard for automated assessment systems.

The system's scalable architecture, configurable evaluation criteria, and comprehensive analytics make it suitable for diverse educational contexts, from individual classrooms to large-scale institutional deployments. With continued development and validation, GradX has the potential to transform educational assessment practices globally, making evaluation more efficient, consistent, and meaningful for both educators and learners.
The successful implementation of GradX will not only reduce faculty workload but also enhance learning outcomes through detailed, constructive feedback, ultimately contributing to improved educational quality and student success.